van Dantzig

G01 / 问题与假设

1 2.1

U

V

$$\mathbb{E}e^{t(U+iV)} = 1 \quad (t \in \mathbb{R}) \quad \Longleftrightarrow \quad \varphi_V(t) = \frac{1}{M_U(t)}.$$

G02 / 两项判据与分布恢复

2.2–2.3 A–B

Fourier V $1/M_U$

$$f_U^*(x) = \frac{1}{\pi}\int_0^\infty \frac{\cos(tx)}{M_U(t)}\,dt \geq 0 \quad (\forall x \in \mathbb{R}).$$

G03 / 矩递推与离散谱恢复

2.3 2.4 B

$u_n = \mathbb{E}U^{2n}$ v_n Hankel N Jacobi V

$$v_0 = 1, \quad v_n = \sum_{k=1}^{n} (-1)^{k+1}\binom{2n}{2k} u_k v_{n-k}, \qquad (v_{i+j})_{i,j=0}^{N},\ (v_{i+j+1})_{i,j=0}^{N} \succeq 0.$$

G04 / 整函数延拓的均值恒等式

2.5 B

$W = U + iV$ F Taylor F f

$$\sum_{n\geq 0} \frac{|F^{(n)}(x)|}{n!}\mathbb{E}|W|^n < \infty, \qquad \mathbb{E}[\operatorname{Re} F(x+W)] = f(x).$$

van Dantzig

G05 / 一般多项式构造

3.1 3.3 C E

P $P(0)=1$ $P(s)>0$ $s\ge 0$ b τ
Laplace Gaussian

$$M_U(t)=e^{\tau t^2/2}(1+bt^2)P(t^2),\qquad \varphi_V(t)=\frac{e^{-\tau t^2/2}}{(1+bt^2)P(t^2)}.$$

G06 / 超出两个比较类的分布族

4 D–E

$a_j\ge 0$ $c_j>0$ φ_U $\mathcal{D}_L$ $\mathcal{D}_P$
Laplace $P(s)=1+s^2$ $b=2$ $\tau=4$
$1/[(1+bt^2)(1+t^4)]$ $b\ge 2$

$$P(s)=\prod_{j=1}^{m}\frac{(s-a_j)^2+c_j^2}{a_j^2+c_j^2},\qquad \varphi_U\in\mathcal{D}\setminus(\mathcal{D}_L\cup\mathcal{D}_P).$$

G07 / 已有分布族与阅读路径

5–6 F

$\mathcal{D}_L$ Gaussian Laplace $\mathcal{D}_P$ Konstantopoulos Patie Sarkar 2024
4.1 Bernstein Gaussian Gaussian Bessel
2 3–4 5 6

$$V_{\mathcal{D}_L}=G+\sum_{n}L_n,\qquad V_{\mathcal{D}_P}=\sqrt{2T}\,Z.$$

Fourier Stieltjes 2–4
C–E